Platform Competition of Temporary Help Services*

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Abstract

We study the platform competition of temporary help service (THS) firms to evaluate their market power. We characterize the THS as a match maker in the two-sided market that benefits from the sizes of worker and client pools. Using the administrative census data of THS establishments in Japan during 2010-2014, we first demonstrate that the THS exercises market power in local markets partly through network effect. We then develop and estimate a model of the THS firm's platform competition. A counterfactual simulation shows that the market power decreases the total surplus by xxx%. Regulating the service fee/wage margin increases the wage and fee and improves the total and worker surplus. Increasing the minimum wage shifts the client firm's labor demand from the outside market to the THS firms and improves the total, THS, and worker surplus. Thus, they are efficient policies to protect temporary workers against the market power of THS firms. These policy experiments highlight the importance of considering the two-sided nature of THS business.

Keywords: Temporary Help Service, Temporary Worker, Two-sided Market, Platform Competition, Market Power, Structural Estimation.
JEL Codes: J20, J22, J23, J31, J38, J42, L10, L11, L13, L40, L84.

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1 Introduction

A non-negligible fraction of workers are employed through the temporary help service (THS): The number amount to 3.4% in the UK, 2.9% in France, 2.0% in Japan and the US in 2018 (World Employment Federation). The THS firm is a two-sided platform that procures labor service from the labor market by paying wages and matches with client firms by charging fees. Thus, the THS firms can potentially exercise market power in both the labor and the service markets, and distort the allocation of labor service. This concern is not merely hypothetical considering the global giants in the THS industry, such as Adeco, Ranstead, Manpower, and Recruit, and leads us to multiple policy relevant questions. Do THS firms exercise the market power in the labor and/or service market? Does the two-sidedness of the THS firm play an important role in the exercise of market power? If so, how large is the welfare loss due to the market power? Should the government regulate the THS? If so, how should the government regulate the operation of the THS firms to improve the efficiency of the market and to protect the workers? What are the implications for the antitrust policy on the THS firms? Addressing these questions is important, because temporary help workers are typically not covered by the traditional labor market institutions to protect workers such as the labor union.

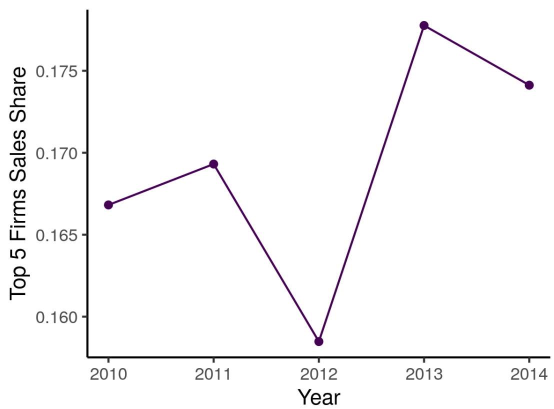
The fact that the THS firms are competing two-sided platforms entails questions on the THS firms' pricing behavior, but the lack of micro data on fees and wages set by the individual THS firms has hampered economists' efforts to analyze their pricing behavior. A strand of literature demonstrates that client firms use THS firms to attain flexible employment adjustment to idiosyncratic product demand shocks (**??**). The other strand of literature analyzes the role of THS as a training institution (**???**). These studies characterize the THS as an institution to attain efficient worker-firm matching and to upgrade workers' skills in the labor market with friction, but do not analyze the THS firms' competition. Recent paper by **?** analyzes the wage determination of THS workers in relation to the direct employees' wages of client firms, but it does not analyze the fee determination and nor relate them to the THS firms' competition.

This study leverages the regulator's administrative census data of the THS establishments in Japan between 2010 and 2014 to analyze their fee and wage setting behaviors. The standard THS establishments operating in Japan must be certified by the regulatory authority, Ministry of Health, Welfare, and Labour. Accordingly, they must submit annual operation reports to the authority. The report includes the fees charged to client firms and wages paid to workers, the number of workers, client firms, and matches, along with the THS establishment's basic characteristics and the zip code level location. TBA: descriptive statistics.

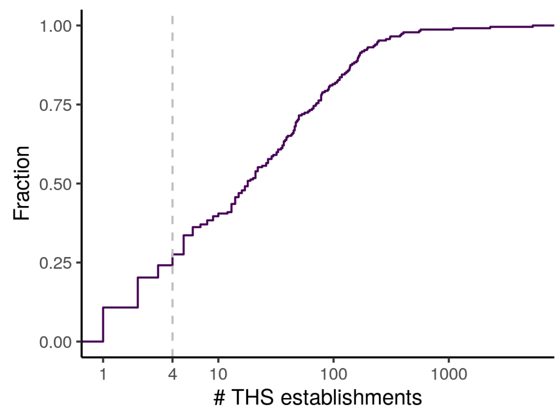
Using the data, we first show that the THS firm can exercise market power in the local geographic market defined by the commuting zone, by demonstrating that the wage and fee changes as the number of THS establishments increases in the market, as predicted from the platform competition model. THS establishments are forced to lower the fee as the number of competitors increases, but maintain the margin by lowering the wage. We next estimate reduced-form labor supply of workers to THS and service demand of clients. The estimates indicate that the supply of workers depends on the wage paid to workers and the service fee charged on clients. Similarly, the demand by the client depends on the service fee charged on clients and the wage paid to workers. These cross elasticities demonstrate the two-sided nature of the THS firm, suggesting the indirect network effects in the operation of THS. Our structural model incorporate this feature.

We next examine how the market power of THS affects the welfare and how government policies, such as fee/wage margin regulation or minimum wage, alter the welfare. For this, we lay out an empirical model of the THS firm's platform competition, where the THS firms set wages and fees, workers and client firms decide which THS firm to use, and the the match is realized according to the THS establishment level matching function. We build our model based on the platform competition model of **?**, which extends the aggregateive games of **?** to the two-sided market. In the model, the expected indirect utility of the worker and client firm depends on the matching probability, and the network effects arise through the effects on the matching probability of the worker and client firm's participation in the THS establishment. The model generates an analytical expression of the worker and client firm THS demand, while allowing for heterogeneous matching efficiency and quality of the THS establishment and network effects. The equilibrium wage and fee of the THS firms are characterized by first-order conditions similar to the standard oligopoly pricing model.

We estimate the model parameters. We first estimate the matching function of the THS establishments. To address the endogeneity of the number of workers and client, we use the mark up instrumental variable that captures the degree of product differentiation by THS establishments **???**. The estimation result shows that the elasticity of the match is 0.566 to the number of workers and 0.360 to the number of client firms. Thus, the match probability is more elastic to the number of workers than to the number of client firms. The constant return to scale is rejected, suggesting that the economy of scale may not work in the THS industry. We next estimate the demand and cost functions. We use the same markup instruments for the demand estimation. The estimation results show that the wage elasticity of worker THS demand is 2.65 and the fee elasticity of client firm THS demand is 4.05, suggesting that client firms are more elastic than workers. The estimation results also confirms the



(a) The National Market



(b) The Local Market

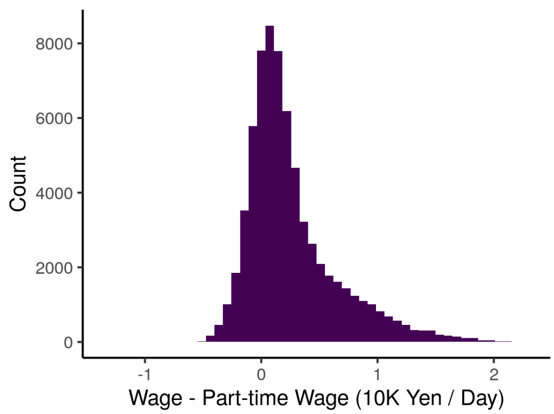
Figure 1: Market Concentration

presence of two externality; the first externality is crowding out effect, an additional worker or client joining to a THS platform reduces the matching probability of incumbent workers or clients. The second externality is the network effect; An additional worker registered to a THS increases the matching probability with a client, and an additional client registered similarly increases the matching probability with a worker. The estimation results show that the negative crowding out effect dominates the positive network effects on both sides of the market, lowering the price elasticity from the pure wage elasticity of 3.46 for workers and 6.04 for client firms. Thus, it is important to take into account the network effects in analyzing the pricing behavior of THS firms.

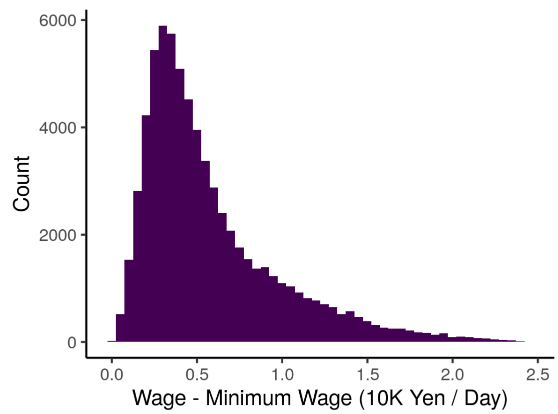
We examine the welfare loss due to the market power THS possesses and THS neglects the network externality. We simulate the market equilibrium where a) THS behaves as a price taker and b) THS internalize the externality based on the structural model estimates,

2 Temporary Help Service

The THS had been prohibited until 1985, when the Worker Dispatching Act was enacted. In 1986, it started for 13 job categories and the duration of the job was limited to 9 months. Since then, the THS industry has experienced a series of deregulations. In 1996, the whitelist was expanded to 26 job categories and the duration was extended to 1 year. In 1999, the THS became available for all jobs except for the blacklist of 5 job categories of manufacturer, construction, port, security, and medical services, and the duration became unlimited except for 26 job categories. In 2004, the manufacturer was dropped from the blacklist, and the duration for the limited 26 job categories was extended to 3 years from 1 year. Thus,



(a) Relative to Part-time Wage



(b) Relative to Minimum Wage

Figure 2: Wage Distribution

it became widely available across the economy. A few changes were introduced in 2012. Importantly, it became compulsory to disclose the average margin on the government website.

In 2014, the employees working for THS firms was 1.17 millions, comprising of 2.2% of the total employees according to the Labor Force Survey. The THS firms that hire employees under a temporal contract to dispatch to the client firms are classified as the standard THS firms. They must be approved by the Ministry of Health, Labour, and Welfare. The approval is given based on asset holding, office floor area, and the ability to manage personal information. The THS firms that hire employees under a permanent contract to dispatch to the client firms are classified as the special operator. They only have to register with the Ministry. In 2015, the category of special THS firms is abolished.

The market share of the top 5 THS firms at the national level is around 16-18% during 2010-2014 as in Figure **??**. However, in the local market ofcommutingg zones, the number of THS firms can be small. Figure **??** shows that the 25% of the commuting zones have no greater than 4 THS establishments. The wages of temporary jobs are relatively low compared to the average wage. However, they tend to be higher than the part-time wages and the minimum wage. Figure **??** shows the distribution of the wage difference to the local part-time wage. Except for a few THS establishments, the wages are higher than the local part-time wages. Figure **??** shows the distribution of the wage difference to the local minimum wage. All temporary job wages are strictly higher than the local minimum wage.

3 Model

We lay out a model of THS establishment competition in a geographic market based on **?**'s oligopoly pricing model with network externality that extends **?**'s aggregative game. The market is two-sided. Workers are on one side of the market and firms are on the other side of the market. The THS establishment can facilitate the matching between workers and firms more efficiently than the outside labor market. The THS establishments compete in the wage paid to the worker and the fee charged to the client firm.

We assume a single-homing for both workers and client firms. Although in reality the workers and client firms register with multiple THS establishments, they cannot match with more than one counterpart at the same time, and the wage and fee are paid only when they match. No wage or fee is paid just for the registration. Therefore, single-homing is more appropriate than multi-homing if we focus on the active workers and client firms in the THS establishment.

There are homogeneous S^W workers and S^F client firms in the market. There is a set of THS establishments $\mathbf{N} = \{1, \ldots, n\}$ in the market. Some THS establishments belong to the same firm. We use $\mathbf{\Delta}$ to describe an ownership matrix where (i, j) element Δ_{ij} takes 1 if THS establishment i and j belongs to the same firm and takes 0 otherwise.

3.1 THS Matching Function

When n_i^W workers and n_i^F client firms use platform i, the number of matches in the platform is determined by the establishment-level matching function

$$Q_i = \mu_i (n_i^W)^{\alpha} (n_i^F)^{\beta}, \tag{1}$$

where μ_i represents the matching efficiency of the THS establishment, and α and β represent the elasticity of matches to the number of registered workers and client firms. For a worker, the probability of a match conditional on using platform i is $\frac{Q_i}{n_i^W} = \mu_i (n_i^W)^{\alpha-1} (n_i^F)^{\beta}$. From a firm, the probability of a match conditional on using platform i is $\frac{Q_i}{n_i^F} = \mu_i (n_i^W)^{\alpha} (n_i^F)^{\beta-1}$.

3.2 THS Demand of Workers and Client Firms

The expected utility for a worker from using THS establishment i offering a wage of w_i is

$$\hat{u}_i = \underbrace{\mu_i (n_i^W)^{\alpha-1} (n_i^F)^{\beta}}_{\text{Probability of match}} \times \underbrace{h_i^W(w_i)}_{\text{Indirect utility conditional on match}} \times \underbrace{\exp(\epsilon_i^W)}_{\text{Taste shock}}, \tag{2}$$

where the mean utility is normalized to zero if the worker is not matched with a client firm. We define

$$u_i = \log \hat{u}_i = \log h_i^W(w_i) + \log \mu_i - (1-\alpha)\log(n_i^W) + \beta \log(n_i^F) + \epsilon_i^W. \tag{3}$$

Workers can choose an outside option that yields a utility of $u_0 = \bar{u}_0 + \epsilon_0^W$. Taste shocks $(\epsilon_0^W, \epsilon_1^W, \ldots, \epsilon_n^W)$ follow the i.i.d. type-I extreme value distribution.

The expected profit for a client firm from using THS establishment i charging a fee of f_i is

$$\hat{v}_i = \underbrace{\mu_i (n_i^W)^\alpha (n_i^F)^{\beta-1}}_{\text{Probability of match}} \times \underbrace{h_i^F(f_i)}_{\text{Indirect utility conditional on match}} \times \underbrace{\exp(\epsilon_i^F)}_{\text{Taste shock}}. \tag{4}$$

We define

$$v_i = \log \hat{v}_i = \log h_i^F(f_i) + \log \mu_i + \alpha \log(n_i^W) + (\beta - 1)\log(n_i^F) + \epsilon_i^F. \tag{5}$$

Firms can choose an outside option that yields $v_0 = \bar{v}_0 + \epsilon_0^F$. Taste shocks $(\epsilon_0^F, \epsilon_1^F, \ldots, \epsilon_n^F)$ follow the i.i.d. type-I extreme value distribution.

Given the number of workers and firms of the THS establishment $\{n_i^W, n_i^F\}_{i \in \mathbf{N}}$ and prices $p = \{w_i, f_i\}_{i \in \mathbf{N}}$, workers and firms pick a THS establishment that yields the highest payoff. Thus, the choice probabilities are

$$s_i^W = \Pr(u_i \geq \max_{i \in \mathbf{N} \cup \{0\}} u_i), \tag{6}$$

$$s_i^F = \Pr(v_i \geq \max_{i \in \mathbf{N} \cup \{0\}} v_i). \tag{7}$$

Since the taste shocks follow an i.i.d. type-I extreme value distribution, we obtain the following closed-form expression for the choice probabilities.

$$s_i^W = \frac{\mu_i h_i^W(w_i)(n_i^W)^{\alpha-1}(n_i^F)^\beta}{\bar{u}_0 + \sum_{j \in \mathbf{N}} \mu_j h_j^W(w_j)(n_j^W)^{\alpha-1}(n_j^F)^\beta}, \tag{8}$$

$$s_i^F = \frac{\mu_i h_i^F(f_i)(n_i^W)^\alpha (n_i^F)^{\beta-1}}{\bar{v}_0 + \sum_{j \in \mathbf{N}} \mu_j h_j^F(f_j)(n_j^W)^\alpha (n_j^F)^{\beta-1}}. \tag{9}$$

The outside option for a worker is to find a part-time job and that for a client firm is to find a part-time job in the outside labor market. We assume that the outside labor market is associated with the matching function that has the same parameters α and β but different matching efficiency μ_0. When the part-time wage is w_0, the mean utilities of the outside options are

$$\bar{u}_0 = \log h_0^W(w_0) + \log \mu_0 - (1-\alpha)\log(n_0^W) + \beta \log(n_0^F), \tag{10}$$
$$\bar{v}_0 = \log h_0^F(w_0) + \log \mu_0 + \alpha \log(n_0^W) - (1-\beta)\log(n_0^F). \tag{11}$$

We assume that the expectations of workers and firms are rational in the sense that the realized number of registered workers and client firms is consistent with the choice probability, that is,

$$n_i^W = s_i^W \times S^W, \tag{12}$$
$$n_i^F = s_i^F \times S^F. \tag{13}$$

Note that n_i^W and n_i^F show up both sides of these equations since s_i^W and s_i^F depend on n_i^W and n_i^F. We derive THS demand functions by solving these equations. Let $\mathbf{p} = (w_1, f_1, w_2, f_2, \ldots, w_N, f_N)$ be a vector of wages and fees. THS demand $D_i^W(\mathbf{p})$ on the worker side and $D_i^F(\mathbf{p})$ on the client firm side are given by the solutions to equations (**??**) and (**??**). Appendix **??** shows that solving those equations yields

$$n_i^W = D_i^W(\mathbf{p}) = S^W \frac{\mu_i^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_i^W(w_i)^{1+\frac{\beta+2(\alpha-1)}{2(2-\alpha-\beta)}} h_i^F(f_i)^{\frac{\beta}{2(2-\alpha-\beta)}}}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_j^W(w_j)^{1+\frac{\beta+2(\alpha-1)}{2(2-\alpha-\beta)}} h_i^F(f_j)^{\frac{\beta}{2(2-\alpha-\beta)}}}, \tag{14}$$
$$n_i^F = D_i^F(\mathbf{p}) = S^F \frac{\mu_i^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_i^F(f_i)^{1+\frac{\alpha+2(\beta-1)}{2(2-\alpha-\beta)}} h_i^W(w_i)^{\frac{\alpha}{2(2-\alpha-\beta)}}}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_j^F(f_j)^{1+\frac{\alpha+2(\beta-1)}{2(2-\alpha-\beta)}} h_i^W(w_j)^{\frac{\alpha}{2(2-\alpha-\beta)}}}. \tag{15}$$

Importantly, THS demand from workers depend on the fee and THS demand from client firms depend on the wage.

3.3 THS Pricing Equilibrium

The profit function of THS establishment i is

$$\Pi_i(\mathbf{p}) = Q_i(\mathbf{p})(f_i - w_i) - D_i^W(\mathbf{p})c_i^W - D_i^F(\mathbf{p})c_i^F, \tag{16}$$

where $Q_i(\mathbf{p}) = \mu_i[D_i^W(\mathbf{p})]^\alpha[D_i^F(\mathbf{p})]^\beta$ is the matches on THS establishment i, c_i^W is a marginal cost of serving an additional worker and c_i^F is a marginal cost of serving an additional client firm. The THS firms simultaneously set prices to maximize the profits of THS establishments belonging to the firm given the prices set by other THS firms. We call the Nash equilibrium of this game as a pricing equilibrium.

The equilibrium prices are given by the solution to the following set of first-order conditions.

$$-Q_i(\mathbf{p}) + \sum_{j\in\mathbf{N}} \Delta_{ij} \left[\frac{\partial Q_j(\mathbf{p})}{\partial w_i}(f_j - w_j) - \frac{\partial D_j^W(\mathbf{p})}{\partial w_i} c_j^W - \frac{\partial D_j^F(\mathbf{p})}{\partial w_i} c_j^F \right] = 0, \tag{17}$$

$$Q_i(\mathbf{p}) + \sum_{j\in\mathbf{N}}^{N} \Delta_{ij} \left[\frac{\partial Q_j(\mathbf{p})}{\partial f_i}(f_j - w_j) - \frac{\partial D_j^W(\mathbf{p})}{\partial f_i} c_j^W - \frac{\partial D_j^F(\mathbf{p})}{\partial f_i} c_j^F \right] = 0, \tag{18}$$

for each establishment $i \in \mathbf{N}$.

? shows that in a similar model, if there is no direct network effect and the indirect network effect is close to zero, the pricing equilibrium exists and unique. If the indirect network effect is not small enough, then there is no guarantee for the uniqueness of the equilibrium. Our model is a special case of **?** except that the profit functions are slightly different. Consumers pay prices for each membership in his model while consumers pay prices only if matches are realized in our model.

4 Estimation

The estimation proceeds in two steps. First, we estimate the establishment matching function using the number of registered workers, client firms, and dispatched workers. Given the estimates of the matching function parameters, we then estimate the preference parameters of workers and client firms and THS cost parameters.

4.1 Estimating Matching Function

Because we have data on THS establishments in multiple markets over time, we now include the subscripts m (commuting zone) and t (year). Taking a log of the matching function (**??**), we have

$$\log Q_{imt} = \alpha \log n_{imt}^W + \beta \log n_{imt}^F + \log \mu_{imt}. \tag{19}$$

We assume that the match efficiency can be decomposed into year fixed effect, commuting zone fixed effect, and the idiosyncratic term.

Instruments Because THS establishments set wages and fees based on the match efficiency, and n_{imt}^W and n_{imt}^F respond to prices, we have an endogeneity problem. We construct two types of instruments for markup (and hence on n_{imt}^W and n_{imt}^F) based on the proximity between THS establishments in the characteristics space (**???**). First, for each establishment i, we compute the distance between establishment i and all the other establishments in the characteristic space, compute the distribution of the distance in the market. Let $d_{i,j}^{(k)} = |x_i^{(k)} - x_j^{(k)}|$ be the distance between establishment i and j in terms of k-th establishment characteristics. We then compute the mean and standard deviation of $\{d_{i,j}^{(k)}\}_{j=1}^N$ for all k and use them as instruments for the wage and fee of establishment i. Second, we count the number of other THS establishments within a certain distance from establishment i. For each establishment i, we divide the distance to other establishments into six bins and count the number of establishments that fall in each of those bins.

4.2 Estimating THS Demand and Cost

THS Demand Specification We specify the mean indirect utility of the match as

$$h_{imt}^W(w_{imt}) = \exp\left[\lambda^W(a_{imt}^W + w_{imt})\right], \tag{20}$$

$$h_{imt}^F(f_{imt}) = \exp\left[\lambda^F(a_{imt}^F - f_{imt})\right], \tag{21}$$

where λ^W and λ^F capture the sensitivity of the utility to the wage and fee. The terms a_{imt}^W and a_{imt}^F capture the quality of the THS establishment. Using these expressions and substituting the network size on each side given by (**??**) and (**??**), the indirect utility on each side can be rewritten as

$$u_i = \beta^{D,W}(a_{imt}^W + w_{imt}) + \beta^{D,F}(a_{imt}^F - f_{imt}) + \left(2 - \frac{1-\alpha-\beta}{2-\alpha-\beta}\right)\log\mu_{imt} + \epsilon_i^W, \tag{22}$$

$$v_i = \beta^{S,W}(a_{imt}^W + w_{imt}) + \beta^{S,F}(a_{imt}^F - f_{imt}) + \left(2 - \frac{1-\alpha-\beta}{2-\alpha-\beta}\right)\log\mu_{imt} + \epsilon_i^W, \tag{23}$$

where the coefficients are represented by a combination of the parameters of the matching function (α, β) and the preference parameters (λ^W, λ^F)

$$\beta^{D,W} = \lambda^W\left(1 + \frac{\beta + 2(\alpha-1)}{2(2-\alpha-\beta)}\right), \tag{24}$$

$$\beta^{S,W} = \lambda^W\frac{\alpha}{2(2-\alpha-\beta)}, \tag{25}$$

$$\beta^{D,F} = \lambda^F\frac{\beta}{2(2-\alpha-\beta)}, \tag{26}$$

$$\beta^{S,F} = \lambda^F\left(1 + \frac{\alpha + 2(\beta-1)}{2(2-\alpha-\beta)}\right). \tag{27}$$

Although the match efficiency μ_i is obtained as the residual in equation (**??**) for THS establishments, we cannot get the match efficiency of the outside labor market μ_0. Since we cannot distinguish between the quality and the match efficiency of the outside labor market, we impose the following normalization

$$\beta^{D,W}a^W_{0mt} + \beta^{D,F}a^F_{0mt} + \left(2 - \frac{1-\alpha-\beta}{2-\alpha-\beta}\right)\log\mu_{0mt} = 0, \tag{28}$$

$$\beta^{S,F}a^F_{0mt} + \beta^{S,W}a^W_{0mt} + \left(2 - \frac{1-\alpha-\beta}{2-\alpha-\beta}\right)\log\mu_{0mt} = 0. \tag{29}$$

Thus, a^W_{imt} and a^F_{imt} capture not only (i) how much establishment i is preferred compared to the outside labor market, but also (ii) how much establishment i has better matching efficiency compared to the outside labor market.

We specify the quality of the THS establishment as

$$a^W_{imt} = x'_{imt}\gamma^{D,W} + \xi^{D,W}_{imt}, \tag{30}$$

$$a^F_{imt} = x'_{imt}\gamma^{D,F} + \xi^{D,F}_{imt}, \tag{31}$$

where x_{imt} is a vector of observed establishment characteristics, and $\xi^{D,W}_{imt}$ and $\xi^{D,F}_{imt}$ are unobserved heterogeneity on the worker and client firm sides.

THS Cost Specification We specify the marginal costs as

$$c^W_{imt} = x_{imt}\gamma^{S,W} + \xi^{S,W}_{imt}, \tag{32}$$

$$c^F_{imt} = x_{imt}\gamma^{S,F} + \xi^{S,F}_{imt}, \tag{33}$$

where $\xi^{S,W}_{imt}$ and $\xi^{S,F}_{imt}$ are unobserved cost shocks.

THS Demand Moments Let us denote the share of establishment i in market m at time t by $s^W_{imt} = \frac{D^W_{imt}}{S^W_{mt}}$ and $s^F_{imt} = \frac{D^F_{imt}}{S^F_{mt}}$. We further define efficiency-adjusted shares as

$$\tilde{s}^J_{imt} = \frac{s^J_{imt}}{\mu_{imt}^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}}}, \quad J = W, F, \tag{34}$$

for $i \neq 0$. Then equations (**??**) and (**??**) imply

$$\log\left(\frac{\tilde{s}^W_{imt}}{s^W_{0mt}}\right) = \beta^{D,W}(w_{imt} - w_{0mt}) - \beta^{D,F}(f_{imt} - w_{0mt}) + x'_{imt}\tilde{\gamma}^{D,W} + \tilde{\xi}^{D,F}_{imt}, \tag{35}$$

$$\log\left(\frac{\tilde{s}^F_{imt}}{s^F_{0mt}}\right) = \beta^{S,W}(w_{imt} - w_{0mt}) - \beta^{S,F}(f_{imt} - w_{0mt}) + x'_{imt}\tilde{\gamma}^{D,F} + \tilde{\xi}^{D,W}_{imt}, \tag{36}$$

where

$$\tilde{\gamma}^{D,W} = \beta^{D,W}\gamma^{D,W} + \beta^{D,F}\gamma^{D,F}, \tag{37}$$

$$\tilde{\gamma}^{D,F} = \beta^{S,W}\gamma^{D,W} + \beta^{S,F}\gamma^{D,F}, \tag{38}$$

$$\tilde{\xi}^{D,W}_{imt} = \beta^{D,W}\xi^{D,W}_{imt} + \beta^{D,F}\xi^{D,F}_{imt}, \tag{39}$$

$$\tilde{\xi}^{D,F}_{imt} = \beta^{S,W}\xi^{D,W}_{imt} + \beta^{S,F}\xi^{D,F}_{imt}. \tag{40}$$

We construct moment conditions

$$\mathbb{E}[\tilde{\xi}^{D,W}_{imt} z_{imt}] = 0, \tag{41}$$

$$\mathbb{E}[\tilde{\xi}^{D,F}_{imt} z_{imt}] = 0, \tag{42}$$

where z_{imt} is a vector of instruments.

THS Cost Moments In addition to the demand side moments, we also use the supply side moments. Using the first-order conditions (**??**) and (**??**), we can back out the marginal costs. The derivation is in Appendix **??**. We then construct the following moment conditions:

$$\mathbb{E}[\xi^{S,W}_{imt} z_{imt}] = 0, \tag{43}$$

$$\mathbb{E}[\xi^{S,F}_{imt} z_{imt}] = 0. \tag{44}$$

Generalized Method of Moments (GMM) Estimator We estimate the parameters by the GMM. Let θ be the vector of model parameters. Let T be the sample size. We first construct the sample moments

$$g(\theta) = \frac{1}{T}\sum_{imt} g_{imt}(\theta) \quad \text{where} \quad g_{imt}(\theta) = \begin{pmatrix} \xi^{D,W}_{imt}(\theta) z_{imt} \\ \xi^{D,F}_{imt}(\theta) z_{imt} \\ \xi^{S,W}_{imt}(\theta) z_{imt} \\ \xi^{S,F}_{imt}(\theta) z_{imt} \end{pmatrix}. \tag{45}$$

We estimate the parameters by minimizing the following GMM objective function

$$g(\theta)' W g(\theta) \tag{46}$$

where W is a weighting matrix. In constructing a weighting matrix, we first set $W = I$ and get estimated parameters $\hat{\theta}$. With $\hat{\theta}$, we obtain an estimate of the efficient weighting matrix

$$\hat{W} = \hat{\Omega}^{-1} \quad \text{where} \quad \hat{\Omega} = \frac{1}{T} \sum_{imt} g_{imt}(\hat{\theta})' g_{imt}(\hat{\theta}). \tag{47}$$

Setting $W = \hat{W}$, we estimate the parameters by minimizing the GMM objective (**??**).

We use the same set of instruments as in the estimation of the matching function, i.e., for establishment i, the mean and the standard deviation of the distance between i and all other establishments, and the number of other establishments within a certain distance in the characteristic space. A vector of covariates x_{imt} contains various characteristics of THS establishment.

5 Data

5.1 Source

Our main data come from the census data of the temporary help service (THS) establishments in Japan from 2010 to 2014. The Ministry of Health, Labour and Welfare collected the data for the administrative purpose. They only disclose the aggregate data to the public,[1] but granted us access to the original data for the research purpose. The data contain detailed information on each establishment, including zip code, average daily wages, average daily fees, annual sales, number of registered workers and client firms, number of matched workers, number of staffs, type of job that the workers engaged in, whether dispatching workers overseas, whether providing temp-to-perm jobs, and whether providing employment placement service. We also use municipality-level aggregate information regarding the size of labor force and the number of part-time or temp workers from the Population Census in 2010 and 2015, and the number of establishments and the number of firms obtained from the Economic Census in 2009 and 2014.

5.2 Data Construction

We construct a panel of THS establishments, but the establishments do not have the identification number in the data. Therefore, when a firm has multiple establishments in the same zip code area, the establishments are not not distinguished.

[1] See `https://www.mhlw.go.jp/stf/seisakunitsuite/bunya/0000079194.html`, accessed on January 31, 2022.

We use the number of registered workers, client firms, and dispatched workers in the data to compute THS demand from workers, THS demand from client firms, and the number of matches, respectively. The number of registered workers in data counts the average number of workers registered on the THS establishment per day but excludes those who have never been actually dispatched. The number of client firms is the number of establishments to which the THS establishment has dispatched workers in the past year. The number of dispatched workers is the average number of dispatched workers per day.

In this paper, we restrict the sample to the standard THS establishment, because the nature of the specialized THS is different from the standard temporary help services. We also discard observations with missing values assuming that they are inactive. We also discard observations reporting zero number of registered workers, temp workers, or clients assuming that they are also inactive establishments. In addition, there are some observations reporting wages or fees that are extremely high or low. We exclude observations reporting wages that are lower than minimum wages or above the 99 percentile, and observation reporting fees that are below the 1 percentile or above the 99 percentile, because the data accuracy is questionable for these observations.

We define a geographic market for the THS establishments based on the commuting zone, because the THS establishments should be competing for the workers living in the same commuting zone whereas not for workers in different commuting zones. **?** studied the commuting pattern of residents and demonstrated that the county is too small and the prefecture is too large as the commuting zone. Thus, they defined the commuting zone by merging multiple counties until the labor supply across commuting zones becomes scarce enough. We use the commuting zone data created by **?**.

When we construct the supply side moments to estimate the THS cost function, we need to recover the marginal costs from the first-order conditions. As we can see in (**??**) in Appendix **??**, recovering marginal costs involves inverting the matrix, the size of which is two times the number of THS establishments in the market. In our data, there are some markets that have, for example, more than 1,000 establishments. Recovering the marginal costs of the establishment of such a market in each iteration of the minimization of the GMM objective is extremely time-consuming. To make the estimation feasible, we exclude the largest five commuting zones from the estimation sample of THS demand and cost.

Since the aggregate information from the Population Census and the Economic Census is municipality-level, we aggregate it to the commuting zone level. The Population Census provides information in the years 2010 and 2015, and the Economic Census provides information in the years 2009 and 2014. We linearly interpolate to get data between those years. They are located in the metropolitan area of Tokyo, Osaka, and Aichi. These markets are

Table 1: Summary Statistics: Market-Level Variables

	Count	Mean	S.D.	Min	Median	Max
Num. THS Establishments	1,095	65	255	0	14	3,847
Labor Force	1,095	289,254	559,559	4,528	127,337	6,184,373
Part-Time / Temp Workers	1,095	73,939	140,055	1,018	31,217	1,411,834
Establishments	1,095	13,180	46,134	121	4,322	640,034
Firms	1,095	8,103	21,158	119	3,154	268,331

Note: Labor Force is the number of persons aged 15 or above who are either working or unemployed.

highly competitive and there are few concerns for market power.

We define the worker-side market size as the number of part-time or temporary workers in the market. In defining the firm-side market size, we multiply the number of establishments in the market with the number of average part-time or temporary workers per establishment, because a client firm can hire multiple part-time or temporary workers.

5.3 Summary Statistics

Market-Level Variables Table **??** reports the summary statistics of market-level variables. Median number of establishments is 14. The distribution is highly skewed, and the average number of establishments is 65, substantially larger than the median. The average size of the labor force, i.e., the number of persons aged 15 or older who are either employed or unemployed, is 292,449, while the average number of persons who are either a part-time worker or temp worker is 74,657. The average number of establishments is 13,304, and the average number of firms is 8,190.

THS Establishment-Level Variables Table **??** shows the summary statics of the THS establishment-level variables. On average, daily wages are 11,300 yen while daily fees are 16,100 yen. The mean of annual sales is about 310 million yen. The average number of temp workers under a fixed-term contract is 33, while that of temp workers under a permanent contract is 35. The average number of temp workers under a daily contract is 4. The average number of registered workers is 166. The average number of client establishments is 59. 27% of the THS establishments provide temp-to-perm offers, which are expected to lead to the direct hiring by the client firm. 1% of the THS establishments dispatch workers to other countries. 61% of the THS establishments provide an employment placement service.

Table 2: Summary Statistics: THS Establishment-Level Variables

	Count	Mean	S.D.	Min	Median	Max
Wage (10K Yen / Day)	49,506	1.09	0.37	0.53	0.98	2.91
Fee (10K Yen / Day)	49,506	1.56	0.58	0.80	1.38	4.53
Sales (10K Yen / Year)	49,506	31014.06	127687.32	0.15	11524.38	7998424.72
Temp Workers (Fixed)	49,506	33	163	0	5	12,063
Temp Workers (Permanent)	49,506	35	119	0	8	6,661
Temp Workers (Daily)	49,506	4	21	0	0	1,328
Registered Workers	49,506	167	680	1	44	51,244
Client Establishments	49,506	59	254	1	16	14,161
Temp to Perm	49,506	0.27	0.45	0.00	0.00	1.00
Oversea	49,506	0.01	0.09	0.00	0.00	1.00
Employment Placement	49,506	0.61	0.49	0.00	1.00	1.00

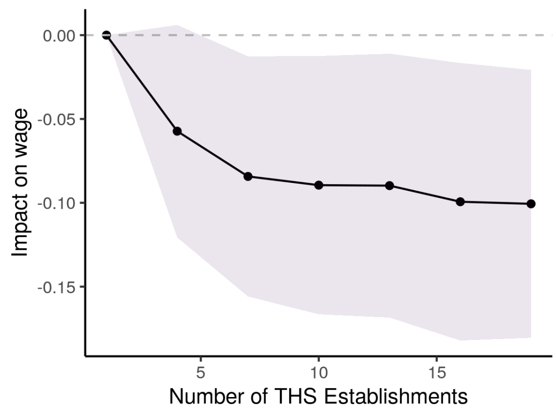
Note: Temp Workers (Fixed) represents the number of temporary workers hired under a fixed-term contract. Temp Workers (Permanent) represents the number of workers hired under a permanent contract. Temp Workers (Daily) represents the number of workers hired under a daily contract. Temp to Perm is an indicator that takes a value of 1 if the establishment provides a temp-to-perm offer, i.e., a temporary position that are expected to lead to a direct hiring by the client firm. Oversea represents an indicator that takes a value of 1 if the establishment dispatches workers to other countries. Employment Placement represents an indicator that takes a value of 1 if the establishment provides an employment placement service.

6 Reduced-Form Analysis

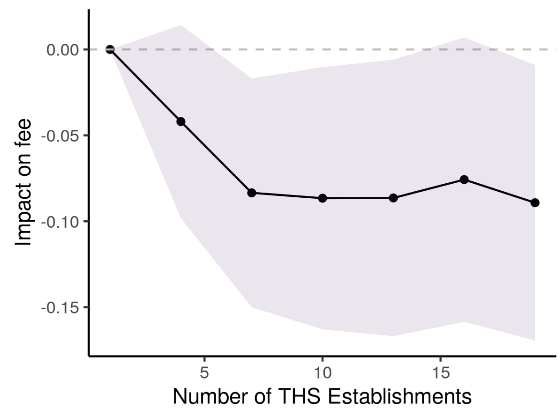
Before demonstrating the estimation results of the structural model, we evaluate the model assumptions using the data. We first evaluate whether the THS establishments exercise some market power and then examine whether there is an indirect network effect at the THS establishment level.

6.1 Competition and Wages, Fees, and Margin

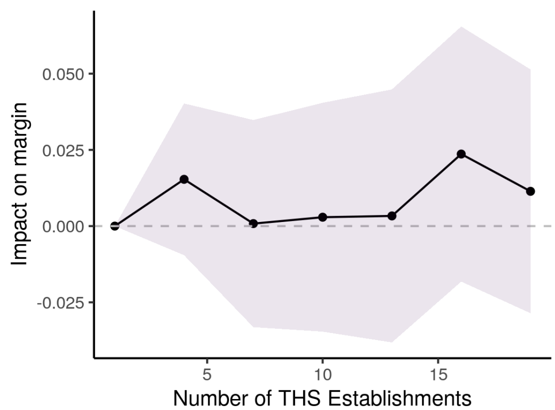
We first demonstrate that the THS establishment's wages and fees respond to the number of establishments in the market. This demonstrates that the THS establishment exercise some market power in wages and fees, because how the price in a market responds to the number of competitors in the market depends on the mode of competition in the market (**???**). For instance, the homogeneous good Cournot model implies that the equilibrium price gradually decreases as the number of competitors increases, whereas the homogeneous good Bertrand model implies that the equilibrium price drops to the competitive level when the number of competitors increases from one to two and it is invariant to the further increase of the number of competitors.



(a) Impact on wages



(b) Impact on fees



(c) Impact on margins

Figure 3: Impact of Competition on Prices

Note: The figure plots the estimated coefficients $\{\hat{\beta}_j\}_{j=2}^{J}$ of the regression equation (**??**) and 90% confidence intervals. The dependent variable is log wages in the left panel, log fees in the right panel, and margins defined as log fees minus log wages in the bottom panel. The standard errors used for constructing the confidence interval is clustered at the commuting zone level.

Specification We conduct a regression analysis in which we regress the wages, fees, or margins of the THS establishments on the degree of competition measured by the number of THS establishments in each market. To flexibly capture the impact of competition on wages, fees, and margins, we regress those outcomes on a set of indicator variables for the number of establishments in a market. Specifically, we estimate the following model

$$y_{imt} = \sum_{j=2}^{J} \beta_j 1[N_{mt} \in \mathbf{N}_j] + \gamma' X_{imt} + \mu_{F(i)} + \lambda_m + \eta_t + \varepsilon_{imt}, \tag{48}$$

where i is the establishment, m is the market, t is the time, and $F(i)$ is the firm that the establishment i belongs to. An outcome variable y_{imt} is the log of wages, the log of fees, or the margin defined as $\log(fee) - \log(wage)$. N_{mt} is the number of establishments in market m at time t. We partition the number of competitors into J blocks. $\mathbf{N}_j$ is a set of integers that need to be specified before the estimation. β_1 is omitted so that β_j for $j = 2, \ldots, J$ represent how large the outcome variables are if $N_{m,t} \in \mathbf{N}_j$ compared to the baseline of $N_{mt} \in \mathbf{N}_1$. We control for time-invariant unobserved heterogeneity at both the firm level and the market level by controlling for firm fixed effects $\mu_{F(i)}$ and market fixed effects λ_m, and we control for year fixed effects η_t as well. In addition, we control for covariates X_{imt}, including both establishment-level characteristics and market-level aggregate variables. Specifically, we control for indicators for whether the establishment providing temp-to-perm offers, whether dispatching workers overseas, whether providing employment placement services, and whether hiring workers with daily contracts as the establishment-level characteristics. We also control for the size of labor force and the number of firms in the market. ε_{imt} is an error term. We do not control for the establishment level fixed effect, because the matching of the THS establishments across years is imperfect. Standard errors are clustered at the commuting zone level.

Instruments There are endogeneity concerns regarding the number of establishments in the market, because higher fees or smaller wages can be more attractive to potential entrants and lead to a larger number of establishments in the market. To deal with this potential endogeneity issue, we exploit the variation in the total number of establishments in the adjacent markets. Specifically, we create a set of indicators for the number of establishments in the adjacent markets and use them as instruments. The number of establishments in the adjacent markets can be regarded as the number of potential entrants to the market predicting the number of actual entrants because the entry cost is lower for the entrants in the adjacent markets. The identification assumption is that the demand shocks to THS services are uncorrelated across markets. This type of instruments are often used in the

Table 3: Estimation Result of the Reduced-Form THS Demand

	Worker	Firm	Worker	Firm
	(1)	(2)	(3)	(4)
Net Wage	2.592**	0.314	4.683**	1.995
	(1.205)	(0.794)	(2.285)	(1.756)
Net Fee	−2.220***	−0.875*	−3.597**	−1.189
	(0.708)	(0.506)	(1.534)	(1.201)
Obs.	49 294	49 294	49 294	49 294
R-sq.	0.705	0.825	0.653	0.810
IV	Dist	Dist	Num Rivals	Num Rivals
Wage elast.	2.919	0.353	5.273	2.248
Fee elast.	-3.581	-1.411	-5.802	-1.919

Note: The table shows the 2SLS estimation result of equations (**??**) and (**??**). Columns (1) and (2) uses the mean and the standard deviation of the distribution of the distance to other establishments in the same market in terms of each characteristic as instruments. Columns (3) and (4) uses the number of other establishments that fall in each of six bins in the characteristic space. The standard errors are clustered at the firm×zipcode level. *p<0.1; **p<0.05; ***p<0.01.

literature for entry and exit analysis [TBA].

Results Figure **??** reports the estimates for the effect of the number of establishments on log wages, log fees, and margins. Figure **??** shows that wages decrease by about 10% as the number of establishments increases to 10 and do not change after that. Figure **??** shows the similar pattern. As a result, we find almost no impact on margins.

6.2 THS Demand and Network Effects

We next estimate THS demand in the reduced form of the structural model. By doing so, we demonstrate that the indirect network effects exist in the THS demand and the THS establishments face the two-sided market.

Specification We estimate the following reduced-form THS demand functions by workers and client firms

$$\log\left(\frac{s^W_{imt}}{s^W_{0mt}}\right) = \beta^{D,W}(w_{imt} - w_{0mt}) + \beta^{D,F}(f_{imt} - w_{0mt}) + x'_{imt}\gamma^W + \varepsilon^W_{imt}, \tag{49}$$

$$\log\left(\frac{s^F_{imt}}{s^F_{0mt}}\right) = \beta^{S,F}(f_{imt} - w_{0mt}) + \beta^{S,W}(w_{imt} - w_{0mt}) + x'_{imt}\gamma^F + \varepsilon^F_{imt}, \tag{50}$$

where s^W_{imt} and s^F_{imt} are the share of THS establishment i in commuting zone m in year t on the worker side and the client firm side, respectively. The option of $i = 0$ corresponds to the outside option. For workers, the outside option is working for a part-time job. For client firms, the outside option is hiring a part-time worker. w_{imt} and f_{imt} are the average wage and fee of establishment i. x_{imt} is a vector of establishment-level characteristics. It also contains the firm fixed effect, commuting zone fixed effect, and year fixed effect.

As shown in Section **??**, the coefficients $\beta^{D,W}, \beta^{D,F}, \beta^{S,F}$ and $\beta^{S,W}$ are functions of the structural parameters including the preference parameters of workers and client firms, and parameters capturing network effects. We allow workers' demand to depend on fees and client firms' demands to depend on wages. The nonzero values of these coefficients indicate the existence of indirect network effects. We use the same set of instrumental variables with the estimation of the structural model.

Results Table **??** shows the estimation results of the THS demand functions (**??**) and (**??**). In columns 1 and 2, we use the distribution of the distance in the characteristic space as the instruments. Column 1 shows that the worker's THS demand positively responds to wages and negatively to fees. Column 2 shows that the client firm's THS demand also positively responds to wages and negatively to fees, although the response to the wage is not statistically significant. Specifically, the worker share relative to the outside option increases by about 26% if the net wage increases by 1,000 yen, while it decreases by 22% if the net fee increases by 1,000 yen. The client firm share relative to the outside option increases by 3% if the net wage increases by 1,000 yen, while it decreases by 8% if the net fee increases by 1,000 yen. The size of the coefficient on the net fee is larger for workers than for client firms.

In columns 3 and 4, we use the number of other establishments within a certain distance as instruments. We find slightly larger impacts with this specification. The worker share relative to the outside option increases by 47% if the net wage increases by 1,000 yen and it decreases by 36% if the net fee increases by 1,000 yen. Both of these effects are statistically significant. The firm share relative to the outside option increases by 23% if the net wage increases by 1,000 yen and it decreases by 14% if the net fee increases by 1,000 yen. However, they are not statistically significant.

The significant effect of the net fee on the worker demand indicates the indirect network effect at the THS establishments: The workers care about the fees because it affects the client firm's THS demand and the match probability in the THS establishment. Although the difference is not statistically significant, it is inconsistent with the standard two-sided market model. The structural model forces the size of the coefficient for the client firm to

Table 4: Estimation Result of the THS Establishment Matching Function

	Log Matches
	(1)
Log Workers	0.566***
	(0.056)
Log Client Firms	0.360***
	(0.052)
Obs.	49 321
R-sq.	0.808
p-val: CRS	0.000

Note: The table shows the estimation result of equation (**??**). The dependent variable is the number of dispatched workers. Log Worker represents the log number of registered persons and Log Client Firm represents the log number of client establishments. p-val: CRS is the p-value of the Wald test of constant returns to scale $\alpha + \beta = 1$. The standard errors are clustered at the firm×zipcode level. *p<0.1; **p<0.05; ***p<0.01.

be greater than for the worker.

7 Estimation Result

7.1 Matching Function

Table **??** shows the estimation result of the matching function of THS establishment. The elasticity of matches to workers is estimated to be $\alpha = 0.566$ and the elasticity of matches to client firms is estimated to be $\beta = 0.360$. This implies that a 1% increase in the number of workers increases the number of matches more than a 1% increase in the number of client firms. Thus, the THS establishment prefers increasing workers to increasing client firms if everything else is equal.

Adding up the estimated coefficients, we have $\alpha + \beta = 0.926$. The two-sided Wald test of the null hypothesis $\alpha + \beta = 1$ is rejected with the p-value < 0.001. This implies that the establishment matching function exhibits decreasing returns to scale. Thus, the THS business is unlikely to lead to a natural monopoly and the merger may not amplify the existing market powers. Although the current model does not allow the heterogeneity of workers and client firms inside the market, the market can be more finely segmented inside the market in reality. If that is the case, the heterogeneity will show up as decreasing returns to scale in the aggregate matching function.

Table 5: Estimation Result of Demand Parameters

	Worker		Client firm	
	Coef.	S.E.	Coef.	S.E.
Wage	4.02	0.15		
Fee			5.83	0.26
Daily	0.32	0.02	0.08	0.01
Temp to Perm	0.14	0.02	0.12	0.01
Oversea	-0.27	0.08	0.20	0.06
Employment Placement	0.26	0.01	-0.01	0.01

Note: The table shows the estimation result of the preference parameters of workers and client firms. The first row represents the estimate of λ^W while the second row represents the estimate of λ^F. The third-sixth rows represent the estimate of $\gamma^{D,W}$ and $\gamma^{D,F}$ in equations (**??**) and (**??**).

Table 6: Estimation Result of Cost Parameters

	Worker		Client firm	
	Coef.	S.E.	Coef.	S.E.
Daily	-0.09	0.02	-0.04	0.01
Temp to Perm	-0.02	0.02	0.04	0.01
Oversea	0.69	0.11	0.09	0.07
Employment Placement	-0.15	0.02	0.00	0.01

Note: This table shows the estimates of $\gamma^{S,W}$ and $\gamma^{S,F}$ in equations (**??**) and (**??**).

7.2 THS Demand and Cost Functions

Table **??** shows the estimation result of the preference parameters of workers and client firms. The key parameters are the coefficients of wages and fees, λ^W and λ^F, in the indirect utility function. The estimation result shows that $\lambda^W = 3.80$ while $\lambda^F = 5.13$. This suggests that the THS demand of client firms is more elastic than that of workers if there is no network effect. Thus, the THS establishment prefers reducing the fee than increasing the wage if everything else is equal.

Table **??** shows the estimation result of the supply-side parameters. It shows that the marginal cost of having a worker is smaller if they offer a daily job, a temp-to-perm job, and an employment placement service, and if they do not offer an overseas job. On the contrary, it is more costly of having a client firm if they offer a daily job, a temp-to-work job, an oversea job, and an employment placement service. Thus, offering a daily job, a temp-to-perm job, and an employment placement service can cut the cost on the worker side but increases on the client firm side, changing the balance of optimal wages and fees.

Table 7: Decomposition of Price Elasticities of THS Demand

	wage	fee	wage	fee
Total	3.15	1.39	1.11	5.63
Price effect	4.08			8.48
Network effect: Direct	-1.20			-3.34
Network effect: Indirect	0.86	1.39	1.11	3.47
Network effect: Interaction	-0.59			-2.99

Note: This table shows the decomposition of elasticities of THS demand from worker / client firm with respect to wages and fees. The decomposition follows equations (**??**). We first calculate these effects for each establishment and then take the simple average.

7.3 Decomposition of Demand Elasticities

The price elasticity of worker THS demand can be decomposed into the direct effect, direct network effect, indirect network effect, and the interaction of the network effects. Let ϵ_w^W be the wage elasticity of worker THS demand. We decompose ϵ_w^W into four parts

$$\varepsilon_w^W = \varepsilon_{w,price}^W + \varepsilon_{w,direct}^W + \varepsilon_{w,indirect}^W + \varepsilon_{w,interaction}^W. \tag{51}$$

The first term represents the elasticity when there is no network effect. This elasticity is obtained by setting $\alpha = 1$ and $\beta = 0$, because the matching probability for workers no longer depends on the number of workers or client firms in the THS establishment with this setting

$$\varepsilon_{w,price}^W = \left.\frac{\partial \log s_i^W}{\partial \log w_i}\right|_{\alpha=1,\beta=0} = \lambda^W w_i(1 - s_i^W). \tag{52}$$

The other three terms constitute the network effects. The second term in (**??**) is obtained by setting $\beta = 0$, because the matching probability depends on the number of workers but not on the client firms in the THS establishment with this setting

$$\varepsilon_{w,direct}^W = \left.\frac{\partial \log s_i^W}{\partial \log w_i}\right|_{\beta=0} = -\lambda^W w_i(1 - s_i^W)\frac{1-\alpha}{2-\alpha}. \tag{53}$$

This term captures the congestion effect and is negative for $\alpha \in (0, 1)$. The indirect network effect is obtained by setting $\alpha = 1$, because the matching probability no longer depends on the workers but depends on the client firms in the THS establishment with this setting

$$\varepsilon_{w,indirect}^W = \left.\frac{\partial \log s_i^W}{\partial \log w_i}\right|_{\alpha=1} = \lambda^W w_i(1 - s_i^W)\frac{\beta}{2(1-\beta)}. \tag{54}$$

This term is positive and larger to the extent of β. The interaction term is a residual given by

$$\varepsilon^W_{w,interaction} = \lambda^W w_i(1 - s^W_i) \left[\frac{\beta + 2(\alpha - 1)}{2(2 - \alpha - \beta)} + \frac{1 - \alpha}{2 - \alpha} - \frac{\beta}{2(1 - \beta)}\right]. \tag{55}$$

We have the same decomposition formula for the client firm THS demand elasticity.

Table **??** shows the decomposed price elasticities. The first and the last columns of the table show that the worker's wage elasticity is 2.94 and the client firm's fee elasticity is 4.98, implying that the THS establishments can exercise market power more on the worker side than on the client firm side. These elasticities do not only reflect the preference parameters λ^W and λ^F but also capture the network effects.

The decomposition in the first column shows that the direct price effect of the worker's wage elasticity is 3.86 and greater than the total worker's wage elasticity. This is because the direct network effect of -1.17 plus the interaction effect of -0.59 dominates the indirect network effect of 0.85. The indirect network effect of worker's wage elasticity arises from the positive indirect network effect of client firm's wage elasticity of 1.02. An increase of wage increases the worker THS demand, which increases the client firm THS demand and further pushes up the worker THS demand.

A similar result is obtained for the client firm's fee elasticity. The decomposition of the last column shows that the direct price effect of the client firm's fee elasticity is 7.46 and greater than the total client firm's fee elasticity. This is because the direct network effect of -2.91 plus the interaction effect of -2.51 dominates the indirect network effect of 2.94. The indirect network effect of client firm's fee elasticity is also due to the positive indirect network effect of worker's fee elasticity 0f 1.25. These decompositions underscore the importance of considering both the direct and indirect network effects to evaluate the THS firm's market power.

8 Policy Analysis

8.1 Setting

To illustrate the implication of the estimated parameters, we consider a simple hypothetical market and conduct counterfactual simulations using the hypothetical market. Specifically, the hypothetical market has two THS establishments that are heterogeneous in terms of the worker-side THS establishment heterogeneity a^W_i, which takes either +1 or -1 standard deviation of the empirical distribution of a^W_i. All other heterogeneities are set equal to the mean of the empirical distribution. The outside wages are set equal to 8,000 yen in all

Table 8: Parameters for the Simulation in a Hypothetical Market

	Description	Value
a^W_{high}	Worker side preference heterogeneity: high	-1.78
a^W_{low}	Worker side preference heterogeneity: low	-3.09
a^F	Clinet firm side preference heterogeneity	-1.58
c^W	Worker side marginal cost	-0.03
c^F	Client firm side marginal cost	-0.35
μ	Match efficiency	1.83
w_0	Part-time wage	0.80
S^W	Worker side market size	1.00
S^F	Client firm side market size	1.00

markets. The market sizes are set equal 1. Table **??** summarizes the parameters used in the simulation for the hypothetical market.

The welfare implications are evaluated in the actual environment. We define the worker surplus for a single worker at the equilibrium price $\mathbf{p}$ by

$$\begin{aligned} ws(\mathbf{p}) &= \frac{1}{\lambda^W}\mathbb{E}\left[\max_{j\in\mathbf{N}\cup\{0\}} u_j\right] \\ &= \frac{1}{\lambda^W}\left[\log\left(\sum_{j\in\mathbf{N}\cup\{0\}} h_i^W(w_i)\times(n_i^W(\mathbf{p}))^{\alpha-1}\times(n_i^F(\mathbf{p}))^{\beta}\right)+\gamma_E\right], \end{aligned} \tag{56}$$

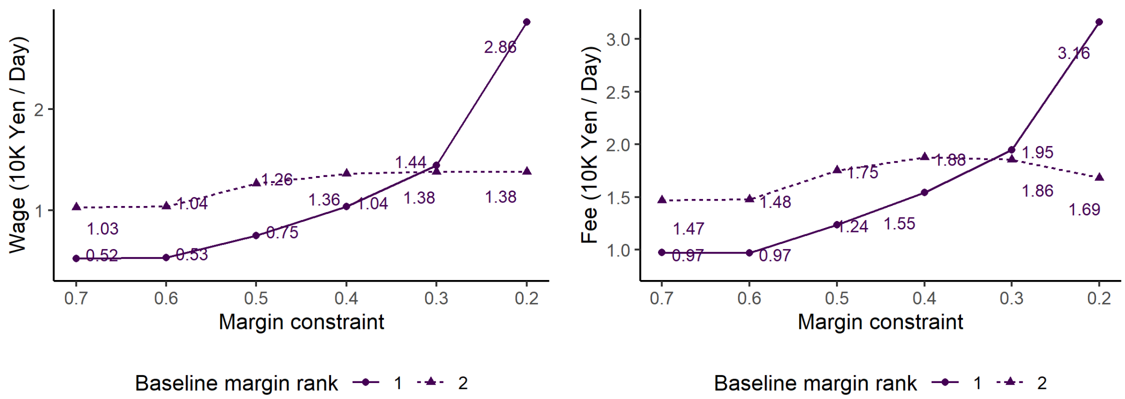
where γ_E is Euler's constant. Similarly, we define the firm surplus for a single client firm at the equilibrium price p by

$$\begin{aligned} fs(\mathbf{p}) &= \frac{1}{\lambda^F}\mathbb{E}\left[\max_{j\in\mathbf{N}\cup\{0\}} v_j\right] \\ &= \frac{1}{\lambda^F}\left[\log\left(\sum_{j\in\mathbf{N}\cup\{0\}} h_i^F(f_i)\times(n_i^W(\mathbf{p}))^{\alpha}\times(n_i^F(\mathbf{p}))^{\beta-1}\right)+\gamma_E\right]. \end{aligned} \tag{57}$$

We obtain the market-level surpluses by multiplying these with the market sizes S^W and S^F, respectively:

$$WS(\mathbf{p}) = S^W \times ws(\mathbf{p}), \tag{58}$$

$$FS(\mathbf{p}) = S^F \times fs(\mathbf{p}). \tag{59}$$



(a) Impact on wages

(b) Impact on fees

Figure 4: Impact of Margin Constraints on Prices: Hypothetical Market

Note: The left panel shows the counterfactual wages for each margin constraint. The right panel shows the counterfactual fees for each margin constraint. There are two establishments in the hypothetical market. The purple line corresponds to the establishment with a larger margin when there is no margin constraint while the yellow line corresponds to the establishment with a lower margin.

8.2 Burden of Market Power

8.3 Margin Constraints

We consider a regulation that imposes an upper bound on the margins of each THS establishment. The Ministry of Health, Labour, and Welfare ordered the THS firms to disclose their margins on the ministry's website in 2012.[2] Even though the ministry did not impose an explicit upper bound on the margin, they meant to put a social pressure on the margins to suppress them.

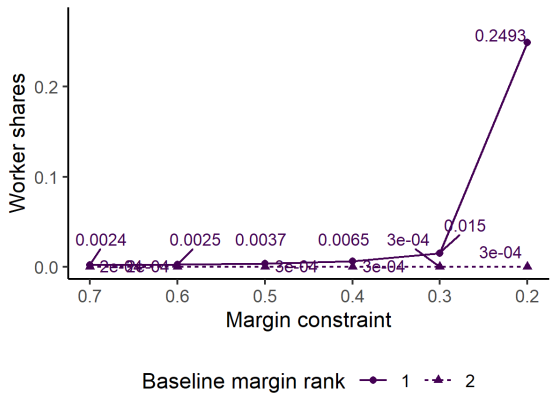
Specifically, we solve for a pricing equilibrium where establishment i solves the following constrained maximization problem given the prices of the other establishments.

$$\max_{w_i, f_i} \sum_{j \in \mathcal{J}_i} \Pi_j(\mathbf{p}) \quad \text{s.t.} \quad \log f_i - \log w_i \leq \bar{m}, \tag{60}$$

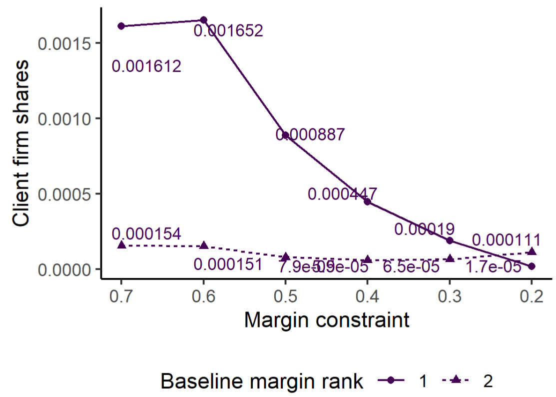
where $\mathcal{J}_i$ is the set of THS establishments that belong to the same firm as establishment i.

The total welfare improves with this regulation as long as no THS firm is forced to exit the market, because it curbs the market power of the THS firms. However, whether it benefits the workers is not clear, because the THS firms can adjust both wages and fees. They may decrease the fee rather than increase the wage, because client firms are more elastic than workers. They may increase the wage rather than decrease the fee, because the matching

[2]https://www.jassa.or.jp/keywords/index5.html, accessed on February 17, 2022.



(a) Impact on worker shares



(b) Impact on client firm shares

Figure 5: Impact of Margin Constraints on Shares: Hypothetical Market

Note: The left panel shows the counterfactual worker shares for each margin constraint. The right panel shows the client firm shares for each margin constraint. There are two establishments in the hypothetical market. The purple line corresponds to the establishment with a larger margin when there is no margin constraint while the yellow line corresponds to the establishment with a lower margin.

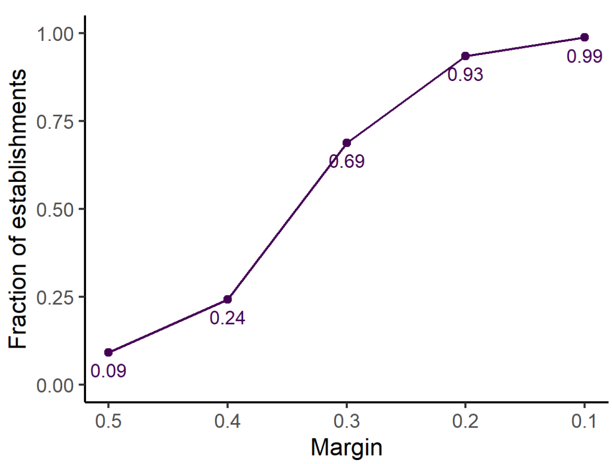
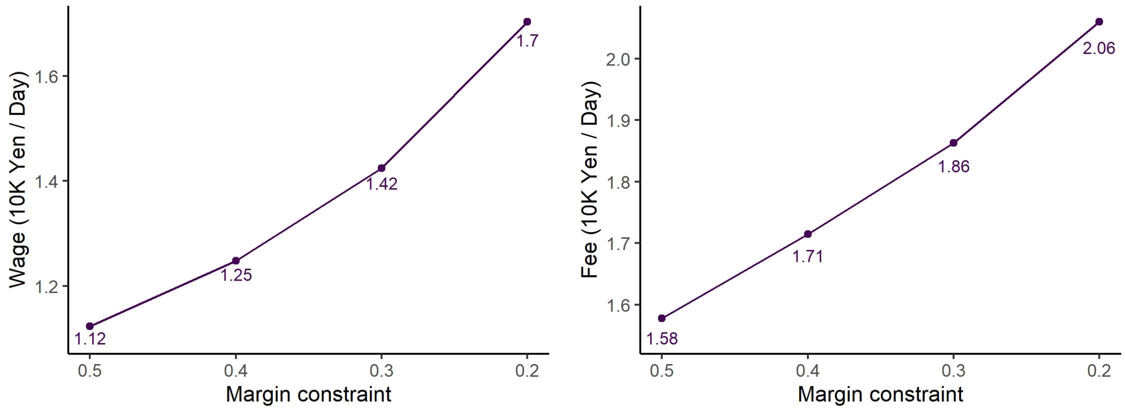


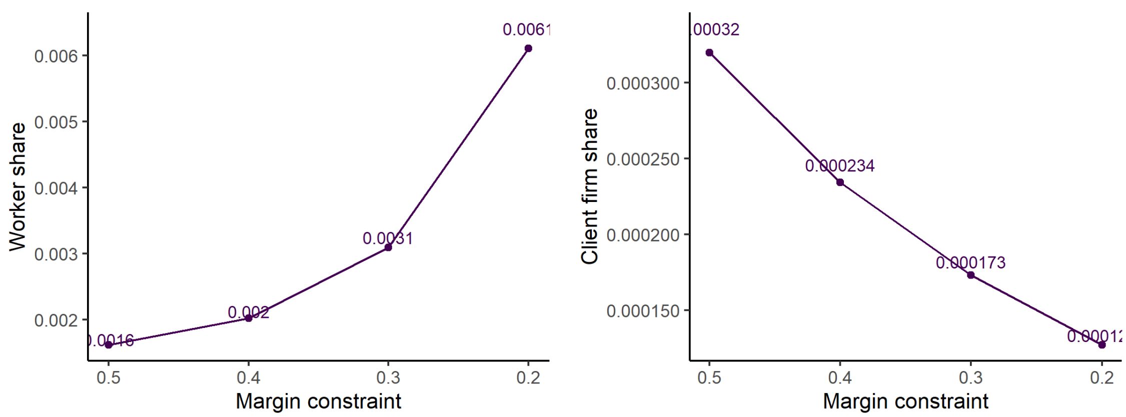
Figure 6: Empirical Distribution of the Margin



(a) Average wage

(b) Average fee

Figure 7: Impact of Margin Constraints on Prices: Actual Markets



(a) Average worker share

(b) Average client firm share

Figure 8: Impact of Margin Constraints on Prices: Actual Markets

Note: This figure shows the impact on the surplus of workers and client firms, and total profits of THS establishments. The calculation of surplus in each market is based on equations (**??**)-(**??**).

Figure 9: Welfare Impact of Margin Constraints

function implies that increasing the number of workers contributes more to the number of matches than increasing the number of client firms.

We solve the system of the following first-order conditions:

$$-Q_i(\mathbf{p}) + \frac{\partial Q_i(\mathbf{p})}{\partial w_i}(f_i - w_i) - \frac{\partial D_i^W(\mathbf{p})}{\partial w_i}c_i^W - \frac{\partial D_i^F(\mathbf{p})}{\partial w_i}c_i^F - \frac{\lambda_i}{w_i} = 0, \tag{61}$$

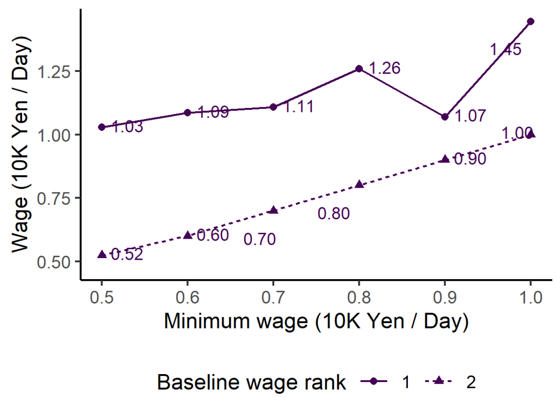
$$Q_i(\mathbf{p}) + \frac{\partial Q_i(\mathbf{p})}{\partial f_i}(f_i - w_i) - \frac{\partial D_i^W(\mathbf{p})}{\partial f_i}c_i^W - \frac{\partial D_i^F(\mathbf{p})}{\partial f_i}c_i^F + \frac{\lambda_i}{f_i} = 0, \tag{62}$$

for $i = 1, 2$ with the complementary slackness condition $\lambda_i[\bar{m} - (\log f_i - \log w_i)] = 0$ where λ_i is the nonnegative Lagrange multiplier.

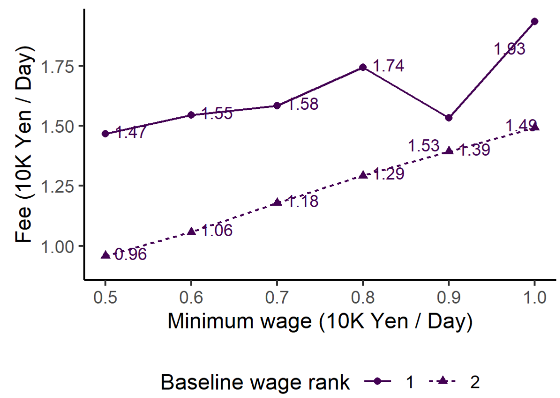
Figure **??** shows how wages and fees respond to the margin constraint. The solid purple line corresponds to the establishment that has a higher quality and higher margin in the baseline equilibrium, while the dashed yellow line corresponds to the establishment that has a lower quality and lower margin in the baseline equilibrium. The left panel shows that both establishments start increasing the wages once they face the binding margin constraint. Likewise, the right panel shows that both establishments start increasing fees. Thus, the THS firms respond to the margin squeeze by increasing the wages and attempt to recover the margin by increasing the fees. Even though the client firms are more elastic, the THS firms increase the wage because the workers contribute more to the matches. Consequently, worker shares increase and client firm shares decrease as the margin constraint becomes more strict as shown in Figure **??**.

Welfare Implications We switch to the actual environment and evaluate the welfare implications of the margin constraint. Figure **??** shows the empirical distribution of the THS establishment margins. This hints the fraction of THS establishments that face the binding margin constraint when the regulation is tightened. 9% of THS establishments set margins that are larger than 0.5 and 7% of THS establishments set margins smaller than 0.2.

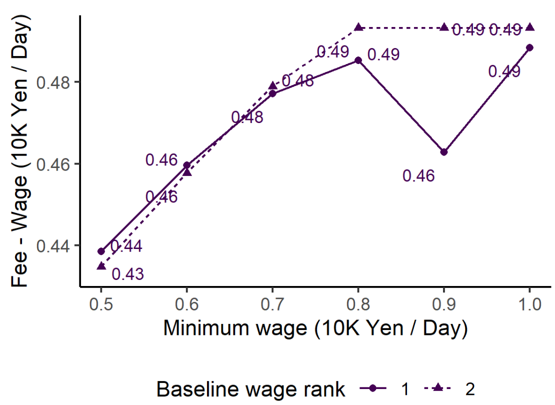
Figures **??** and **??** show the average prices and shares in the simulation using the actual markets. Consistent with the results of the hypothetical market, the average wage increases and the average fee decreases as the margin constraint is tightened. This leads to the increase in the average worker share and the decrease in the average client firm share. Figure **??** shows that the average worker surplus increases and the average client firm surplus decreases due to the margin constraint.



(a) Wage



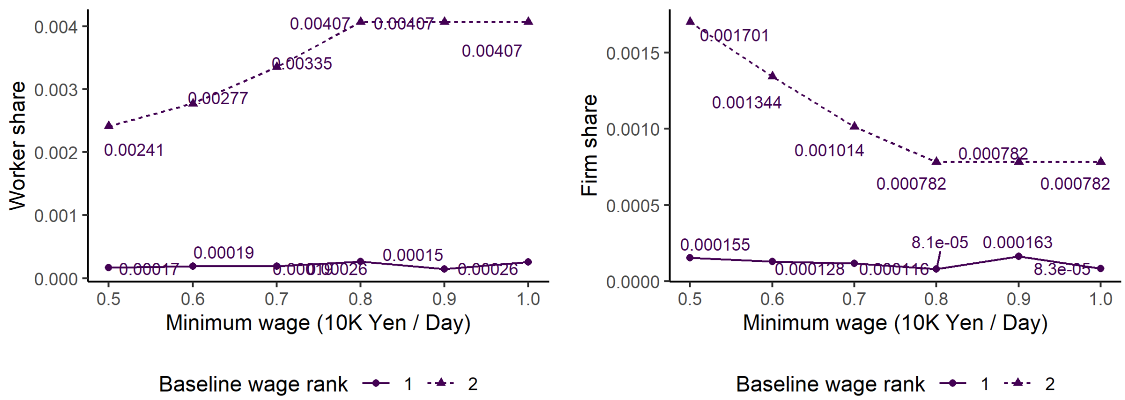
(b) Fee



(c) Fee - wage

Figure 10: Impact of Minimum Wages on Prices

Note: The left panel shows the counterfactual wages for each minimum wage. The right panel shows the counterfactual fees for each minimum wage. There are two establishments in the hypothetical market. The purple solid line corresponds to the establishment with a larger margin when there is no margin constraint while the yellow dashed line corresponds to the establishment with a lower margin.



(a) Impact on wages

(b) Impact on fees

Figure 11: Impact of Minimum Wages on Shares

Note: The left panel shows the counterfactual worker shares for each minimum wage. The right panel shows the counterfactual client firm shares for each minimum wage. There are two establishments in the hypothetical market. The purple solid line corresponds to the establishment with a larger margin when there is no margin constraint while the yellow dashed line corresponds to the establishment with a lower margin.

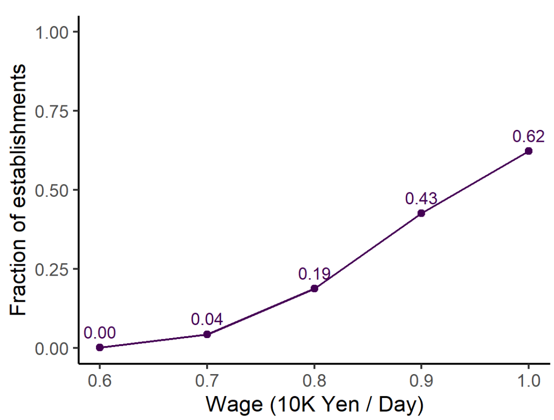
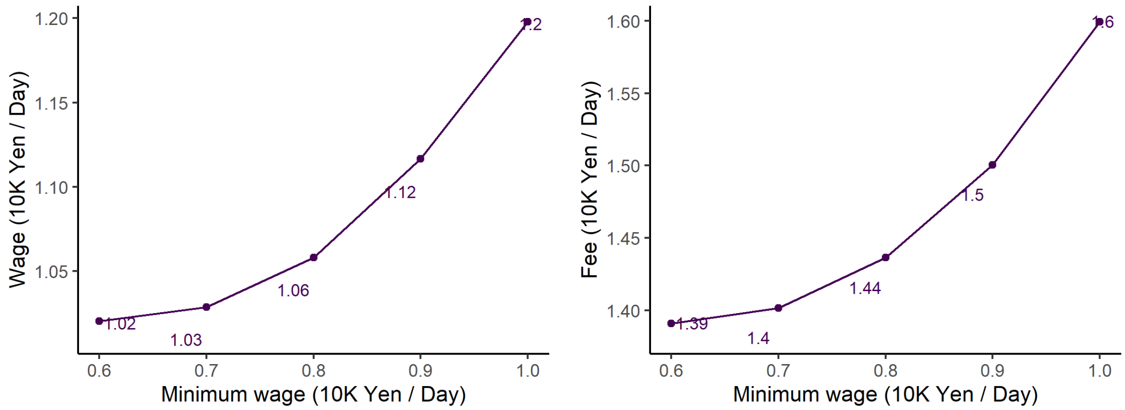


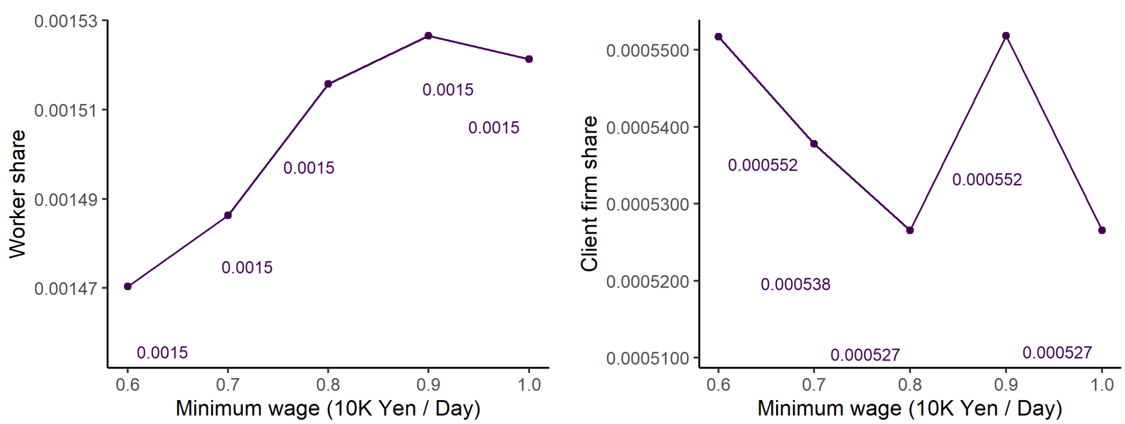
Figure 12: Empirical Distribution of the Wage



(a) Average wage

(b) Average fee

Figure 13: Impact of Minimum Wages on Prices: Actual Markets



(a) Average worker share

(b) Average client firm share

Figure 14: Impact of Minimum Wages on Prices: Actual Markets

Figure 15: Welfare Impact of Minimum Wages

Note: This figure shows the impact on the surplus of workers and client firms, and total profits of THS establishments. The calculation of surplus in each market is based on equations (??)-(??).

8.4 Minimum Wages

We next consider the policy of changing the minimum wage. TBA: actual regulation.

We solve for a pricing equilibrium where establishment i solves the following constrained maximization problem given the prices of the other establishments.

$$\max_{w_i, f_i} \sum_{j \in \mathcal{J}_i} \Pi_j(\mathbf{p}) \quad \text{s.t.} \quad w_i \geq \underline{w}, \tag{63}$$

where $\mathcal{J}_i$ is the set of THS establishments that belong to the same firm as establishment i. We replace the outside wages by $\underline{w}$ if the observed outside wage is smaller than the minimum wage.

We solve the system of the following first-order conditions

$$-Q_i(\mathbf{p}) + \frac{\partial Q_i(\mathbf{p})}{\partial w_i}(f_i - w_i) - \frac{\partial D_i^W(\mathbf{p})}{\partial w_i} c_i^W - \frac{\partial D_i^F(\mathbf{p})}{\partial w_i} c_i^F - \lambda_i = 0, \tag{64}$$

$$Q_i(\mathbf{p}) + \frac{\partial Q_i(\mathbf{p})}{\partial f_i}(f_i - w_i) - \frac{\partial D_i^W(\mathbf{p})}{\partial f_i} c_i^W - \frac{\partial D_i^F(\mathbf{p})}{\partial f_i} c_i^F = 0, \tag{65}$$

for $i = 1, 2$ with the complementary slackness condition $\lambda_i(\underline{w} - w_i) = 0$ where λ_i is the nonnegative Lagrange multiplier.

The increase of the minimum wage increases the wage the THS firm must pay if their wages are restricted by the minimum wage. However, the increase of minimum wage also increases the wage the client firms must pay when they hire part-time workers in the outside labor market. This induces the shift of the client firm's demand from the outside labor market to the THS firms, benefiting the THS firms. Because the part-time wage of the outside labor market is lower than most of the temporary worker's wages, the effect of the client firm demand substitution will dominate under a moderate increase of the minimum wage. The increase of unemployment is often concerned when increasing the minimum wage. Because our model incorporates the unemployment due to the mismatch, we can also evaluate the effect on the unemployment of the minimum wage increase.

Figure **??**(a) and (b) show how wages and fees respond to the minimum wage increase. The solid purple line corresponds to the establishment that has a higher quality and a higher wage in the baseline equilibrium, and the dashed yellow line corresponds to the establishment that has a lower quality and a lower wage in the baseline equilibrium. The low-wage establishment is bound by the minimum wage even when the minimum wage is low, and it keeps increasing both the wage and the fee in accordance with the minimum wage increase. The high-wage establishment starts increasing the wage and fee once the minimum wage hits 8,000 yen. At the minimum wage of 8,000 yen, the high-wage establishment is

Table 9: Relevant Market Test

	Wage	Fee	Wage or Fee
Fraction of markets with $\Delta\Pi^M > 0$	0.532	0.466	0.536

Note: The table reports the fraction of markets that can be profitably monopolized. The first column shows the fraction of markets where a hypothetical monopolist can increase profits by decreasing wages so that margin of each establishment goes up by 5% from the observed margin. The second column shows the case where hypothetical monopolists increase margins by increasing fees. The third column shows the fraction of markets where a hypothetical monopolist can increase profits either by increasing fees or decreasing wages.

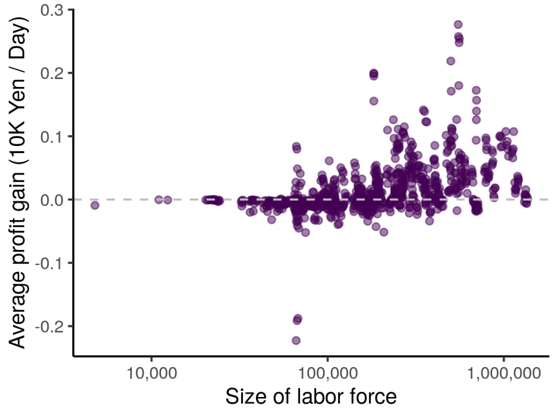
not yet bound by the minimum wage, but it increases the wage to compete with the outside option.

Figure **??** shows the changes in the margin due to the minimum wage increase. It shows that the low-wage establishment can increase the margin due to the minimum wage increase, but the high-wage establishment keeps the margin. The low-wage establishment stops increasing the margin when the minimum wage reaches 8,000 yen. TBA: What is going on?

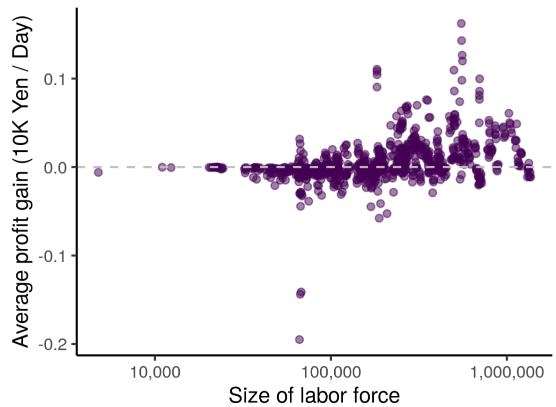
Welfare Impact We switch to the actual environment and evaluate the welfare implications of the minimum wage increase. Figure **??** shows the empirical distribution of the THS establishment level wage. It shows the fraction of THS establishments that will face a binding minimum wage constraint at each value of the minimum wage. 4% of the establishments face the constraint when the minimum wage is set at 7,000 yen per day. When the minimum wage is set at 8,000 yen, it binds for 19% of the establishments, and when the minimum wage is set 10,000 yen, it binds for 62% of establishments. Figure **??** show that both the average wage and fee increases as the minimum wage increases, and Figure **??** show that both the worker and client firm shares increase. Thus, the increase of the minimum wage can benefit the THS establishments by shifting the client firm share from the outside labor market to the THS establishments. TBA: unemployment. Figure **??** shows the impact of the minimum increase wage on the surplus of workers and client firms and the total profit of THS establishments. TBA: Describe numbers.

8.5 Hypothetical Monopolist Test

Antitrust authorities are often required to define a relevant market when they evaluate the market power of firms. A common approach taken in merger reviews assessing product markets is the small significant non-transitory increase in price (SSNIP) test. The SSNIP test asks whether a hypothetical monopolist of a set of products can profitably increase



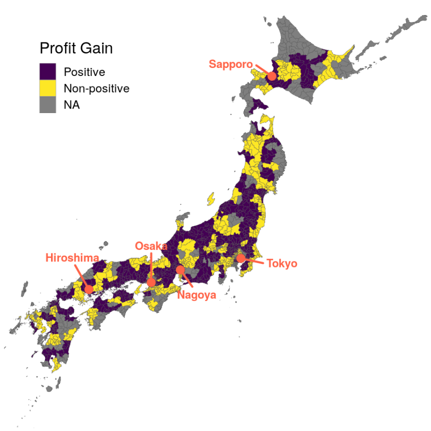
(a) Wage increase



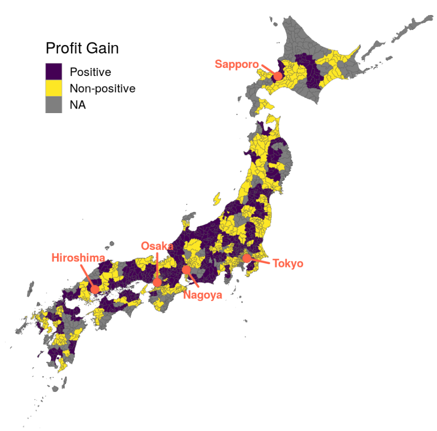
(b) Fee increase

Note: This figure shows the relationship between the market size represented by the size of labor force in each market and the average profit gain of establishment in each market. Each point corresponds to each commuting zone in each year.

Figure 16: Relevant Market Test: Average Profit Gain vs. Market Size



(a) Wage increase



(b) Fee increase

Note: This figure shows the commuting zones in which a hypothetical monopolist experiences a profit gain by either decreasing wages or increasing fees. For each commuting zone, profits are averaged over time.

Figure 17: Relevant Market Test: Geographic Distribution

the price by a few percentages, typically 5%. If the hypothetical monopolist can profitably increase the price, then there are few substitutes other than the set of products. Then, we conclude that the set of products constitutes an antitrust relevant market.

Defining a relevant market for the THS firms is not straightforward. First, it is not clear to what degree the hiring through the THS firms can be separated from the hiring in the labor market. One extreme is to consider the entire labor market and the other extreme is to consider only the hiring through the THS firms. Second, the geographic market must be taken into account, because the labor market is highly segmented across geographic locations. Even if the THS industry is competitive at the national level, there can be some local geographic markets where the THS firms can exercise market power. Third, the two-sidedness of the THS firms requires the correction of the standard hypothetical monopolist test, because the THS firms can change either of the wage and fee.

We test whether the THS establishment of each commuting zone can be regarded as the antitrust relevant market. Based on the estimated model, we increase the margins of the THS establishments in a commuting zone by 5%, either by decreasing the wage or by increasing the fee. If this increases the joint profit of the THS establishments, then they can be regarded as the relevant market, because the hypothetical monopolist can optimize the combination of the wage cut and the fee increase to further increase the profit. Therefore, this procedure results in a conservative definition of the relevant market.

Formally, for each commuting zone, we consider the profit of a hypothetical monopolist

$$\Pi^M(\mathbf{p}) = \sum_{i=1}^{N} \Pi_i(\mathbf{p}), \tag{66}$$

where $i = 1, \ldots, N$ are THS establishments in the commuting zone. Let $\mathbf{p}^*$ be the price profile in the observed equilibrium. First, we consider an alternative price profile $\mathbf{p}^W$ where fees are set equal to the observed equilibrium while wages are chosen to increase the margin by 5%. Second, we consider another price profit $\mathbf{p}^F$ where wages are set equal to the observed equilibrium while fees are chosen to increase the margin by 5%. We calculate $\Delta\Pi^M(\mathbf{p}^J) = \Pi^M(\mathbf{p}^J) - \Pi^M(\mathbf{p}^*)$ for $J = W, F$ for each commuting zone.

Table **??** reports the fraction of markets where we find an increase in the profit of a hypothetical monopolist when we decrease wages or increase fees. We find that the profits of hypothetical monopolists increase in 53.2% of markets when we decrease wages, while they increase in 46.6% of markets when we increase fees. The profits of hypothetical monopolists increase in 53.6% of markets due to either the decrease in wages or the increase in fees.

Figure **??** shows the relationship between the market size and the profit gain in each market. We use the size of labor force as the market size. The profit gain in each market is the average profit gain per establishment in the market. The figures show that hypothetical monopolists are more likely to gain profits in larger markets.

Figure **??** shows the geographic distribution of commuting zones in which a hypothetical monopolist experiences a profit gain.

9 Concluding Remarks

This paper studied the platform competition of the Japanese THS firms and evaluated the implications of their market power using the administrative census data of the THS establishments. We demonstrated that the THS firms could exercise the market power, and that the indirect network effects exist in the worker and client firm THS demand. We developed and estimated the model of THS firm competition and conducted various policy experiments based on the estimated model. The policy experiments showed the importance of considering the two-sidedness in the analysis of the THS firms' pricing behavior. We found the regulations such as constraining the margin and increasing the minimum wage could increase the worker surplus by curbing the market power of the THS firms. Importantly, they could improve the total surplus. The paper showed that the structural analysis of labor market institutions can be used to evaluate the welfare implications of labor market regulations and policies. It also offered tools for antitrust policies on the labor market.

There are several limitations in this paper. First, we did not consider the training of early career workers, which has been regarded as one of the most important roles of the THS firms in the literature. If we curb the market power of the THS firms by regulation, they may cut the investment in early career workers, worsening the worker welfare. For this analysis, we need to augment the information about the training activity with the current THS data. Second, we did not allow heterogeneity in the worker and client firm THS demand, and did not consider the horizontal differentiation of the THS establishment to derive the analytical solution to the worker and client firm THS demand. As the decreasing-return-to-scale matching function suggests, the market for the THS may be more finely segmented inside the commuting zone. Third, TBA. Finally, TBA.

A Appendix

A.1 Derivation of Network Sizes

First, we have

$$\frac{n_i^W}{n_j^W} = \frac{\mu_i h_i^W(w_i)(n_i^W)^{\alpha-1}(n_i^F)^{\beta}}{\mu_j h_j^W(w_j)(n_j^W)^{\alpha-1}(n_j^F)^{\beta}} \tag{67}$$

$$= \left[\frac{\mu_i h_i^W(w_i)(n_i^F)^{\beta}}{\mu_j h_j^W(w_j)(n_j^F)^{\beta}}\right]^{\frac{1}{2-\alpha}}. \tag{68}$$

Likewise, we have

$$\frac{n_i^F}{n_j^F} = \frac{\mu_i h_i^F(f_i)(n_i^W)^{\alpha}(n_i^F)^{\beta-1}}{\mu_j h_j^F(f_j)(n_j^W)^{\alpha}(n_j^F)^{\beta-1}} \tag{69}$$

$$= \left[\frac{\mu_i h_i^F(f_i)(n_i^W)^{\alpha}}{\mu_j h_j^F(f_j)(n_j^W)^{\alpha}}\right]^{\frac{1}{2-\beta}}. \tag{70}$$

Combining these, we obtain

$$\frac{n_i^W}{n_j^W} = \left[\frac{\mu_i h_i^W(w_i)(n_i^F)^{\beta}}{\mu_j h_j^W(w_j)(n_j^F)^{\beta}}\right]^{\frac{1}{2-\alpha}} \tag{71}$$

$$= \left[\frac{\mu_i h_i^W(w_i)}{\mu_j h_j^W(w_j)}\left[\frac{\mu_i h_i^F(f_i)(n_i^W)^{\alpha}}{\mu_j h_j^F(f_j)(n_j^W)^{\alpha}}\right]^{\frac{\beta}{2-\beta}}\right]^{\frac{1}{2-\alpha}} \tag{72}$$

$$= \left(\frac{\mu_i}{\mu_j}\right)^{\frac{1}{2-\alpha-\beta}}\left[\frac{h_i^W(w_i)}{h_j^W(w_j)}\left(\frac{h_i^F(f_i)}{h_j^F(f_j)}\right)^{\frac{\beta}{2-\beta}}\right]^{\frac{2-\beta}{2(2-\alpha-\beta)}}. \tag{73}$$

Similar calculation yields

$$\frac{n_i^F}{n_j^F} = \left(\frac{\mu_i}{\mu_j}\right)^{\frac{1}{2-\alpha-\beta}}\left[\frac{h_i^F(f_i)}{h_j^F(f_j)}\left(\frac{h_i^W(w_i)}{h_j^W(w_j)}\right)^{\frac{\alpha}{2-\alpha}}\right]^{\frac{2-\alpha}{2(2-\alpha-\beta)}}. \tag{74}$$

Substituting those expressions into (10) and (11) yields

$$
\begin{aligned}
n_i^W &= S^W \frac{\mu_i h_i^W(w_i)(n_i^W)^{\alpha-1}(n_i^F)^{\beta}}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j h_j^W(w_j)(n_j^W)^{\alpha-1}(n_j^F)^{\beta}} && (75)\\
&= S^W \frac{\mu_i h_i^W(w_i)}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j h_j^W(w_j)\left(\frac{n_j^W}{n_i^W}\right)^{\alpha-1}\left(\frac{n_j^F}{n_i^F}\right)^{\beta}} && (76)\\
&= S^W \frac{\mu_i h_i^W(w_i)}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j h_j^W(w_j)\left(\frac{\mu_i}{\mu_j}\right)^{\frac{\alpha+\beta-1}{2-\alpha-\beta}}\left[\frac{h_j^W(w_j)}{h_i^W(w_i)}\left(\frac{h_j^F(f_j)}{h_i^F(f_i)}\right)^{\frac{\beta}{2-\beta}}\right]^{\frac{(\alpha-1)(2-\beta)}{2(2-\alpha-\beta)}}\left[\frac{h_j^F(f_j)}{h_i^F(f_i)}\left(\frac{h_j^W(w_j)}{h_i^W(w_i)}\right)^{\frac{\alpha}{2-\alpha}}\right]^{\frac{(2-\alpha)\beta}{2(2-\alpha-\beta)}}} && (77)\\
&= S^W \frac{\mu_i^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_i^W(w_i)^{1+\frac{\beta+2(\alpha-1)}{2(2-\alpha-\beta)}} h_i^F(f_i)^{\frac{\beta}{2(2-\alpha-\beta)}}}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_j^W(w_j)^{1+\frac{\beta+2(\alpha-1)}{2(2-\alpha-\beta)}} h_i^F(f_j)^{\frac{\beta}{2(2-\alpha-\beta)}}}, && (78)
\end{aligned}
$$

and

$$
n_i^F = S^F \frac{\mu_i^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_i^F(f_i)^{1+\frac{\alpha+2(\beta-1)}{2(2-\alpha-\beta)}} h_i^W(w_i)^{\frac{\alpha}{2(2-\alpha-\beta)}}}{\sum_{j\in\mathbf{N}\cup\{0\}} \mu_j^{2-\frac{1-\alpha-\beta}{2-\alpha-\beta}} h_j^F(f_j)^{1+\frac{\alpha+2(\beta-1)}{2(2-\alpha-\beta)}} h_i^W(w_j)^{\frac{\alpha}{2(2-\alpha-\beta)}}}. \tag{79}
$$

A.2 Derivation of the Marginal Costs

Profit function of establishment i:

$$
\Pi(\mathbf{p}) = Q_i(\mathbf{p})(f_i - w_i) - D_i^W(\mathbf{p})c_i^W - D_i^F(\mathbf{p})c_i^F, \tag{80}
$$

Consider a firm with establishments $\mathcal{J}'$. FOC with respect to w_j is

$$
\begin{aligned}
0 &= \sum_{i\in\mathcal{J}'} \frac{\partial \Pi_i}{\partial w_j}\\
&= -Q_j + \sum_{i\in\mathcal{J}'}\left[\frac{\partial Q_i}{\partial w_j}(f_i - w_i) - \frac{\partial D_i^W}{\partial w_j}c_i^W - \frac{\partial D_i^F}{\partial w_j}c_i^F\right] && (81)\\
&= -Q_j + \sum_{i\in\mathcal{J}} \Delta_{ij}\left[\frac{\partial Q_i}{\partial w_j}(f_i - w_i) - \frac{\partial D_i^W}{\partial w_j}c_i^W - \frac{\partial D_i^F}{\partial w_j}c_i^F\right]
\end{aligned}
$$

where $\mathcal{J}$ is the set of all establishments in the market, and Δ_{ij} is 1 if i and j are held by the same firm and it is 0 otherwise.

Let $\mathbf{\Delta} = (\Delta_{ij})_{i,j\in\mathcal{J}}$ be the ownership matrix. Let

$$\mathbf{G}_w^Q = \left(\frac{\partial Q_i}{\partial w_j}\right)_{i,j\in\mathcal{J}}, \tag{82}$$

$$\mathbf{G}_w^{DW} = \left(\frac{\partial D_i^W}{\partial w_j}\right)_{i,j\in\mathcal{J}}, \tag{83}$$

$$\mathbf{G}_w^{DF} = \left(\frac{\partial D_i^F}{\partial w_j}\right)_{i,j\in\mathcal{J}}. \tag{84}$$

We get a matrix representation for FOCs for $(w_j)_{j\in\mathcal{J}}$:

$$-\mathbf{Q} + (\mathbf{\Delta} \odot \mathbf{G}_w^Q)(\mathbf{f} - \mathbf{w}) - (\mathbf{\Delta} \odot \mathbf{G}_w^{DW})\mathbf{c}^W - (\mathbf{\Delta} \odot \mathbf{G}_w^{DF})\mathbf{c}^F = 0, \tag{85}$$

where $\odot$ represents an element-wise product. Similarly, we get a matrix representation for FOCs for $(f_j)_{j\in\mathcal{J}}$:

$$\mathbf{Q} + (\mathbf{\Delta} \odot \mathbf{G}_f^Q)(\mathbf{f} - \mathbf{w}) - (\mathbf{\Delta} \odot \mathbf{G}_f^{DW})\mathbf{c}^W - (\mathbf{\Delta} \odot \mathbf{G}_f^{DF})\mathbf{c}^F = 0. \tag{86}$$

The marginal costs can be obtained by

$$\begin{pmatrix} \mathbf{c}^W \\ \mathbf{c}^F \end{pmatrix} = \begin{pmatrix} \mathbf{\Delta} \odot \mathbf{G}_w^{DW} & \mathbf{\Delta} \odot \mathbf{G}_w^{DF} \\ \mathbf{\Delta} \odot \mathbf{G}_f^{DW} & \mathbf{\Delta} \odot \mathbf{G}_f^{DF} \end{pmatrix}^{-1} \begin{pmatrix} -\mathbf{Q} + (\mathbf{\Delta} \odot \mathbf{G}_w^Q)(\mathbf{f} - \mathbf{w}) \\ \mathbf{Q} + (\mathbf{\Delta} \odot \mathbf{G}_f^Q)(\mathbf{f} - \mathbf{w}) \end{pmatrix} \tag{87}$$

A.3 Surplus of Workers and Client Firms

We have

$$ws(p) = \frac{1}{\lambda^W}\left[\log\left(\sum_{j\in\mathbb{N}\cup\{0\}} h_i^W(w_i) \times [n_i^W(p)]^{\alpha-1} \times [n_i^F(p)]^{\beta}\right) + \gamma_E\right]. \tag{88}$$

Substituting the representation of n_i^W and n_i^F given by equations (**??**) and (**??**) yields

$$ws(p) = \frac{1}{\lambda^W}\left[(2-\alpha)H^W(p) - \beta H^F(p) + \gamma_E\right], \tag{89}$$

where

$$H^W(p) = \sum_{j \in \mathbf{N} \cup \{0\}} h_j^W(w_j)^{1+\frac{\beta+2(\alpha-1)}{2(2-\alpha-\beta)}} h_i^F(f_j)^{\frac{\beta}{2(2-\alpha-\beta)}}, \tag{90}$$

$$H^F(p) = \sum_{j \in \mathbf{N} \cup \{0\}} h_j^F(f_j)^{1+\frac{\alpha+2(\beta-1)}{2(2-\alpha-\beta)}} h_i^W(w_j)^{\frac{\alpha}{2(2-\alpha-\beta)}}. \tag{91}$$

Similarly, we have

$$fs(p) = \frac{1}{\lambda^F}\left[(2-\beta)H^F(p) - \alpha H^W(p) + \gamma_E\right]. \tag{92}$$